

Bag of words

Document	Text	
D1	students watch youtube	1
D2	youtube watch youtube	1
D3	students write comments	0
D4	youtube write comments	0
D5	youtube write comment	1

	students	watch	youtube	write	comments
D1	1	1	1	0	0
D2	0	1	1	0	0
D3	1	0	0	1	1
D4	0	0	1	1	1
D5	0	0	1	1	1

$D_1 \rightarrow [1 \ 1 \ 1 \ 0 \ 0]$

$D_2 \rightarrow [0 \ 1 \ 1 \ 0 \ 0]$

$D_3 \rightarrow [1 \ 0 \ 0 \ 1 \ 1]$

$D_4 \rightarrow [0 \ 0 \ 1 \ 1 \ 1]$

$D_5 \rightarrow [0 \ 0 \ 1 \ 1 \ 1]$

✓

Feature	One-Hot Encoding	Bag of Words (BoW)
Applied To	Individual words	Entire documents
Output	Vector for each word	Vector for each document
Captures Frequency?	✗ No	✓ Yes
Vocabulary Required?	✓ Yes	✓ Yes
Sparse?	✓ Yes	✓ Yes
Semantic Meaning?	✗ No	✗ No

text classification

Spam Detection Task

We have emails:

D1: Win money now

D2: Claim your prize

D3: Meeting at 10 AM

Labels:

D1 → Spam

D2 → Spam

D3 → Not Spam

The model needs to learn:

Document → Label

not

Word → Label

Problem with One-Hot Encoding

Vocabulary:

win, money, now, claim, your, prize, meeting, at, 10, am

One-hot vectors:

win → [1,0,0,0,0,0,0,0,0,0]

money → [0,1,0,0,0,0,0,0,0,0]

Now consider:

Win money now

This document contains **3 different vectors**.

Which one should we feed into Logistic Regression?

win vector?

money vector?

now vector?

The classifier expects **one fixed-size vector per sample**, but One-Hot gives multiple vectors for one document.

Bag of Words Solves This

Vocabulary:

win, money, now, claim, your, prize, meeting, at, 10, am

For

Win money now

BoW vector:

[1,1,1,0,0,0,0,0,0,0]

For

Meeting at 10 AM

BoW vector:

[0,0,0,0,0,0,1,1,1,1]

Now every document becomes:

One document → One vector

Perfect for classification.

What the Classifier Learns

Suppose after training:

Word	Weight
win	+4
money	+3
prize	+5
meeting	-4

The model learns:

win + money + prize

usually means **Spam**.

BoW provides the word counts needed to learn these patterns.

Problem

- 1) Sparsity
- 2) size mismatch
- 3) OOV →
- 4) Semantic meaning
 - ↳ I like this movie
 - ↳ I do not like this movie

